

Response Letter with Annotations

Than, N., Fan, L., Law, T., Nelson, L. K., & McCall, L. (2025). Updating “The Future of Coding”: Qualitative Coding with Generative Large Language Models. *Sociological Methods & Research*, 54(3), 849–888. <https://doi.org/10.1177/00491241251339188>

Annotations

1. Purposes

- a. Motivation: to test the ability of generative LLMs to replicate and augment traditional qualitative textual analysis, and propose a workflow for conducting qualitative coding with generative LLMs
- b. Specific Rationale
 - Using deep learning techniques combined with natural language interfaces, compared with using manual engineering of text into numerical representations, has the potential to transform the qualitative coding process, when validated and used with care
 - The LLM results on accuracy are mixed, and there are currently no widely agreed-upon guidelines on how best to use LLMs for qualitative coding practices that involve labeling, classification, annotation, or discovery tasks
- c. Research Questions and/or Hypotheses
 - **RQ:** Can generative LLMs replicate and augment traditional qualitative textual analysis? Which model performs the best under which contexts?
 - **IV:** LLM types: Close-source LLM (ChatGPT-4) vs. Open-source LLMs (Llama3, Llama3.1, Gemma2)
 - **DV:** accuracy (precision, recall, F1 scores)
 - **Contextual factors: prompt strategy**
 1. Prompt structure: zero-shot (no examples) vs. few-shot (providing 1-3 examples)
 2. Definition provided: no definition vs. research generated definition vs. LLM generated definition
 - **“Control variable”:** model temperature (all set to 0)

2. Method

- a. Design: “Experiments” & content analysis
 - Based on a human-coded news report dataset, the paper compares the content classification of different LLMs using various prompt strategies
 - Conditions: LLM types (ChatGPT, Llama3, Llama3.1, Gemma2) * prompt strategy (zero-shot vs. few shot) * definition provision (no definition vs. research-generated definition vs. LLM-generated definition)
 - Prompt structure

Table 2. Core Prompt Structure for Zero-Shot Prompts.

Zero-shot Prompts

Panel A: Relevant vs. irrelevant articles (categories 1-3 vs. category 4; $N = 1,253$)

```
"role": "system",
"content": "You are a news classifier."
"role": "user",
"content": "We categorize articles that are related to issues of
income inequality, changes in income or wealth, general
economic conditions."
"role": "user",
"content": "Read this definition: "+irrelevant_definitiona,
"role": "user",
"content": "Read this article: "+text,
"role": "user",
"content": "Is the article relevant? Answer relevant or
irrelevant, and why in 1 sentence."
```

Panel B: Inequality vs. non-inequality among relevant articles (category 1 vs. categories 2-3, $N = 786$)

```
"role": "system",
"content": "You are a news classifier."
"role": "user",
"content": "We categorize articles that are related to issues of
income inequality, changes in income or wealth, general
economic conditions."
"role": "user",
"content": "Read this definition: "+inequality_definition,
"role": "user",
"content": "Read this article: "+text,
"role": "user",
"content": "Does the article reference American economic
inequality? Respond with 'Yes' if article meets any or all
criteria referencing American economic inequality and 'No' if
article meets none of the criteria, and explain why in 1
sentence."b
```

Notes: ^a As discussed in the text, we provide a definition of irrelevant rather than relevant articles.

^b This command is provided when researcher- and LLM-generated definitions are included, otherwise the command is shorter when no definition is provided.

- Task: let LLMs classify the corpus of articles into the binary classification schemes
- Classification steps:
 1. Whether the articles are related to inequality (1/0)
 2. Whether the article about inequality is about economic inequality (1), economic conditions (2), changes in wages and income (3)

Table 1. Categories and Binary Schemas From Hand-Coding Guide and Analysis.

Categories and Binary Schemas	Number
Categories (shorthand in hand-coding guide) ^a :	
Relevant Articles	
Inequality ^b (relinequality)	1
Economic Conditions (releconomy)	2
Changes in Wages and Income (relchanges)	3
Irrelevant Articles (irrelevant)	4
Binary Classification Schemas:	
Relevant versus Irrelevant	1/0
Inequality versus Other Relevant	1/0

Notes: ^a In the hand-coding guide from the original study (McCall 2013), these categories are given shorthands which are referenced in some of our prompts because the guide itself is evaluated by the LLMs. ^b This category is broken down into two categories (explicit and implicit inequality) in the original work and in the replication but is simplified in most of the present work to be a single category (except for tests in Appendix G in the online supplement, Table G1, Panel B, Rows 1 and 2).

b. Sample

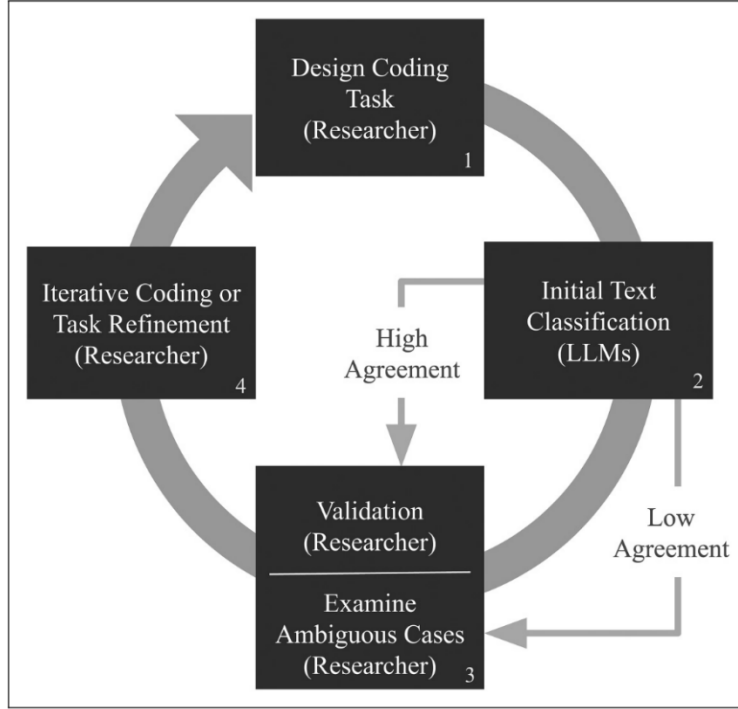
- Human-coded dataset: a sample of 1,253 newsweekly articles appearing in Time, Newsweek, and U.S. News & World Report between 1980 and 2012. The task is about the identification of socio-economic inequality and its manifestation

c. Measures

- Inter-model agreement
 1. Inter-prompt agreement
 2. Inter-model agreement
- Accuracy metrics:
 1. Precision
 2. Recall
 3. F1 score

3. Results and Conclusions

a. Researcher-LLM-Researcher workflow



b. Performance

- According to the F1 score, providing a definition of inequality did not consistently improve performance
- When researcher definition was provided, ChatGPT performed the best across different conditions, and was far superior in few-shots condition
- the zero-shot prompts performed better than the few-shot learning prompts
- LLM-generated definition performed marginally better across all models except GPT4

Table 3. Accuracy Metrics for Three Classification Tests (rows) by LLM and Definition Input (columns).

	No Definition				Researcher Definition				LLM-Generated Definition			
	Llama3 (70b)	Llama3.1 (70b)	Gemma2 (27b)	GPT4	Llama3 (70b)	Llama3.1 (70b)	Gemma2 (27b)	GPT4	Llama3 (70b)	Llama3.1 (70b)	Gemma2 (27b)	GPT4
Relevant:	P: 0.74	P: 0.80	P: 0.78	P: 0.82	P: 0.76	P: 0.79	P: 0.82	P: 0.83	P: 0.82	P: 0.79	P: 0.79	P: 0.81
Zero Shot	R: 0.73	R: 0.81	R: 0.79	R: 0.84	R: 0.74	R: 0.78	R: 0.83	R: 0.83	R: 0.82	R: 0.82	R: 0.83	R: 0.83
(n = 1253)	F1: 0.74	F1: 0.80	F1: 0.79	F1: 0.83	F1: 0.73	F1: 0.78	F1: 0.82	F1: 0.83	F1: 0.82	F1: 0.80	F1: 0.80	F1: 0.81
Inequality:	P: 0.71	P: 0.72	P: 0.73	P: 0.75	P: 0.74	P: 0.72	P: 0.74	P: 0.77	P: 0.76	P: 0.73	P: 0.75	P: 0.74
Zero Shot	R: 0.72	R: 0.72	R: 0.73	R: 0.74	R: 0.75	R: 0.72	R: 0.74	R: 0.77	R: 0.77	R: 0.73	R: 0.75	R: 0.74
(n = 786)	F1: 0.70	F1: 0.72	F1: 0.73	F1: 0.74	F1: 0.73	F1: 0.71	F1: 0.74	F1: 0.77	F1: 0.75	F1: 0.73	F1: 0.75	F1: 0.73
Inequality:	P: 0.72	P: 0.73	P: 0.61	P: 0.75	P: 0.69	P: 0.64	P: 0.49	P: 0.75	–	–	–	–
Few Shot	R: 0.73	R: 0.75	R: 0.77	R: 0.75	R: 0.72	R: 0.77	R: 0.94	R: 0.75				
(n = 786)	F1: 0.73	F1: 0.70	F1: 0.60	F1: 0.75	F1: 0.69	F1: 0.62	F1: 0.59	F1: 0.75				

Note: Average weighted precision (P), recall (R), and F1 metrics across all tests. F1 is the harmonic mean of recall and precision. All tests use the core prompt structure presented in Table 2.

c. Agreement: Including GPT4 in these calculations produced very similar metrics (results not reported)

Table E1. Fleiss' kappa for two classification tests.

	Intermodel Agreement			Interprompt Agreement		
	No Def	Res Def	LLM Def	Llama3	Llama3.1	Gemma2
Relevant: zero-shot (n=1253)	FK = 0.74 PA = 0.84	FK = 0.68 PA = 0.76	FK = 0.63 PA = 0.76	FK = 0.65 PA = 0.74	FK = 0.62 PA = 0.74	FK = 0.68 PA = 0.79
Inequality: zero-shot (n=786)	FK = 0.67 PA = 0.75	FK = 0.67 PA = 0.75	FK = 0.69 PA = 0.76	FK = 0.83 PA = 0.88	FK = 0.78 PA = 0.84	FK = 0.80 PA = 0.85

Note: Fleiss's Kappa (FK) within models across three prompts (Interprompt Agreement, columns 1-3) and within prompts across three open-source models (Intermodel Agreement, columns 4-6). Percent agree (PA) indicates the percent of newspaper articles that received the same classification across all three models using the same prompt (columns 1-3) or all three prompts using the same model (columns 4-6).

- d. Authors' Conclusions:
- LLMs can achieve classifications comparable to hand coding and supervised machine learning
 - although the full GPT4 model achieved marginally better accuracy metrics almost across the board, these metrics were not substantially better than the smaller, pre-quantized, open-source models

Table 4. High-Level Summary of Findings.

Guideline	Key Insight	Why It Matters	Practical Application
Researcher-LLM-Researcher Workflow	Use LLMs as part of an iterative process, with researchers designing tasks, LLMs making an initial pass, and humans validating/refining results.	Ensures LLMs support, rather than replace, qualitative judgment, and redirects researcher efforts in strategic ways.	Use LLMs for first-pass coding and focus researcher effort on validating clear cases and analyzing ambiguous cases.
Leverage Interprompt and Intermodel Agreement	Compare multiple prompts and models to improve consistency and highlight ambiguities.	Reduces risks from biases and/or inaccuracies in any single LLM or prompt.	Identify stable classifications by comparing different prompt structures and models, targeting the human-validation step.
Prioritize Reproducibility and Transparency	Closed models may perform slightly better, but adopting open-source options offers transparency, reproducibility, and accessibility.	Promotes research integrity, long-term usability, and accessibility.	Favor open-source and smaller models, unless accuracy trade-offs are particularly critical.

4. Discussion

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a. Implications

- LLMs may eventually be able to serve as reliable coding assistants, but that they could also offer scalability (e.g., analyzing larger datasets), flexibility (e.g., easily modifying classification criteria and building on pre-trained models), and efficiency (e.g., dramatically reducing or strategically redirecting the need for hand coding).

b. Limitations

- Difficult task for generative LLMs: long document covering multiple themes, may introduce noise into the classification task
- We did not investigate whether our corpus was in fact part of the training data of the models we used, but it should be possible to do so when using open-source models
- We did not engage in extensive fine-tuning tests with all the models employed in our main zero-shot and few-shot tests

c. Future Directions

- How LLMs can be used for different kinds of interpretive text analysis tasks, such as whether an LLM adopting varying political personas would label vignettes about inequality as being normatively good or bad
- It is not yet clear which type of fine-tuning is best for the types of tasks and data we are examining. Future research should assess when, and how much, fine-tuning might be needed for different kinds of models

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Response Letter (~600 words)

Assessment: Than et al. (2025) provide a timely evaluation of LLMs' capability to conduct qualitative analysis on long texts. As LLMs are increasingly integrated into content analysis, this study addresses the critical question of how these models can be applied to qualitative coding and evaluates their actual efficacy. A significant contribution of the paper is giving a tool-performance guide and offering practical suggestions on prompt structures and model selection.

The main take-aways of this paper are particularly inspiring for researchers: for long-text analysis, providing comprehensive coding schemes is often sufficient for both closed-source and open-source models. While GPT-4o offers consistent and high performance, the authors demonstrate that open-source models (e.g., Llama 3) can achieve similar results by generating their own definitions based on the coding book. Crucially, the study finds that providing extra human-verified examples (few-shot prompting) may be unnecessary or even detrimental for long texts, as they often act as semantic noise or exceed token limits. Furthermore, the findings suggest that the labor-intensive and time-consuming process of fine-tuning is often not required. In terms of prompt engineering, assigning a specific role to the LLM (e.g., "you are a news classifier") and placing the command at the end of the prompt are necessary for maximizing coding accuracy.

Weakness: Despite these insights, the study's limitations raise important questions. First, because the experiments do not exhaust every possible condition of model types, prompt structures, and task categories, we must be cautious about declaring which specific model, prompt, and task combination is definitively "best".

This problem also raises another more fundamental concern. The study illustrates how LLM performance fluctuates across different contexts and is influenced by numerous factors. Since these models update rapidly, the effectiveness of a specific model and prompt structure, which is the main contribution of this study, might shift within months. Does this mean we must conduct these kind of large-scale evaluations periodically, just like commercial annual reports? If so, is this a sustainable or meaningful pursuit for the field? Furthermore, we should also be cautious about the interpretation of performance "differences". In the context of understanding complex qualitative data, is a 0.02 difference in an F1 score truly consequential? Instead of focusing on marginal gains in indicators, should our attention shift toward the stability of these models as reliable research assistants?

Finally, I am truly confused about the evolving role of human researchers in this workflow. While the authors argue that humans remain essential in designing coding guidelines, according to the Researcher-LLM-Researcher loop, is the complete outsourcing of text immersion a positive long-term trend? From my limited understanding of qualitative studies, in qualitative research, the researcher is traditionally seen as the primary research tool. Much of the growth, insight, and even the happiness of doing research come from the time-consuming and often painful process of immersing oneself in the text. It is through repeated deliberation and deep interpretation that the "aha moment" which is the instance where the narrative suddenly makes sense, occurs. This sense of serendipity is deeply subjective. I also believe that different researchers may arrive at different findings by engaging with the same material. If this interpretative journey is outsourced to an LLM, does the analytical process lose its intellectual

vitality? Could this lead to findings that are technically accurate but ultimately mediocre, lacking the depth that only a human tool can provide?